Newer Oral Anticoagulants

16

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Clinical Vignettes

Clinical Vignette 1: A 28-year-old woman is 6 weeks postpartum and breastfeeding. She develops left leg swelling and is diagnosed with deep vein thrombosis (DVT). She asks if she can take one of the new oral anticoagulants that she has heard about instead of taking warfarin.

Clinical Vignette 2: Your patient with atrial fibrillation has been on warfarin for many years and has consistently had difficulty maintaining a therapeutic international normalized ratio (INR). He admits to missing doses occasionally despite being advised of the importance of being compliant with dosing instructions. During his office visit he asks about switching to one of the new medications which he has heard about on television.

Clinical Vignette 3: Your patient with mild renal impairment (creatinine clearance of $>50 \leq 80$ ml/min) on dabigatran for atrial fibrillation is going to have surgery and therefore has held his medication for 48 h. Upon admission to the hospital the surgeon is concerned about surgical and postoperative bleeding risks. What laboratory test can be run prior to surgery to ensure that no residual anticoagulant effect remains?

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Dabigatran Etexilate

Dabigatran etexilate is administered orally as a prodrug and converted to the active compound, dabigatran. It is a competitive reversible direct inhibitor of both free and clot bound thrombin. It has a US FDA approval for prevention of stroke and systemic embolization in patients with nonvalvular atrial fibrillation. It has an additional indication in Canada and the UK for VTE prevention after elective hip and knee surgery. A predictable pharmacokinetic profile allows for fixed dosing with no need for coagulation monitoring or dose adjustments.

Key Clinical Trials

Dabigatran was compared directly to adjusted dose warfarin in patients with atrial fibrillation. In a noninferiority trial enrolling 18,113 patients with atrial fibrillation and CHADS2 score (Congestive heart failure, Hypertension, Age ≥75 years, Diabetes mellitus, Stroke, or otherwise history of arterial thromboembolism) of two or more, patients were randomized to receive dabigatran (either 110 or 150 mg twice daily) or adjusted dose warfarin with a median follow-up of 2 years and primary outcome of stroke or systemic embolism (Connolly et al. 2009). The rate of stroke and systemic embolism was similar between warfarin and the 110 mg dabigatran dose with a lower rate of major hemorrhage in

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patients receiving 110 mg of dabigatran. There was a lower rate of stroke and systemic embolism in patients receiving the 150 mg dose of dabigatran (superior) compared to warfarin with similar rates of major hemorrhage. At either dose of dabigatran, the rate of intracranial hemorrhage was 1/3 of the rate in patients taking warfarin.

Dabigatran has been studied in three large phase 3 trials for prevention of VTE following hip or knee surgery altogether randomizing more than 8,000 patients to dabigatran versus enoxaparin (RE-MODEL and RE-MOBILIZE after knee surgery, RE-NOVATE after hip surgery). Patients were randomized to receive either dabigatran (150 or 220 mg once daily) vs. enoxaparin (40 mg once daily or 30 mg twice daily). The primary endpoint in all trials was total VTE events, and a composite of venographic DVT, symptomatic DVT, and all-cause mortality. The duration of thromboprophylaxis was 6–10 days in RE-MODEL, 12–15 days in RE-MOBILIZE, and 28–35 days in RE-NOVATE. In a meta-analysis of all three trials the higher dose dabigatran (220 mg once daily) was noninferior to enoxaparin (40 mg once daily) with a similar rate of major bleeding. Interestingly both doses of dabigatran did not achieve noninferiority compared to enoxaparin (30 mg twice daily) in the RE-MOBILIZE trial. This data led to approval of dabigatran 220 mg once daily for thromboprophylaxis following hip or knee surgery in Europe and Canada (Eriksson et al. 2007a, b; Ginsberg et al. 2009; Wolowacz et al. 2009).

Dabigatran (150 mg twice daily) has been studied for the treatment of acute VTE; acute proximal DVT or pulmonary embolism (PE) or both in phase 3 clinical trials (RE-COVER, RE-COVER II) compared with adjusted dose warfarin (Schulman et al. 2009). Data from the first RE-COVER study group is published. The study was a double-blind, noninferiority design randomizing 2,539 patients to either treatment arm for a 6 month course of therapy. An initial 5–10 day course of parental anticoagulation (either unfractionated heparin or low-molecular-weight heparin) was administered to all patients. The primary endpoints were composite recurrent VTE and VTE-related mortality. Dabigatran was found to be noninferior to warfarin with a similar safety profile in terms of major bleeding risk.

To date, data from RE-COVER 2 has only been presented in abstract form. This study again compared fixed dose dabigatran directly with adjusted dose warfarin for the treatment of acute VTE for 6 months and confirmed noninferiority of dabigatran compared to adjusted dose warfarin with a lower risk of clinically relevant bleeding (Schulman 2012). Additional phase 3 studies have been conducted evaluating the efficacy and safety of extending anticoagulant therapy with dabigatran compared to warfarin (RE-MEDY) or placebo (RE-SONATE). Both studies were double blind and randomized, comparing dabigatran 150 mg twice daily to either warfarin (active-control study) or placebo (placebo-control study). All patients had completed at least a 3 month course of anticoagulation for symptomatic proximal VTE. The active-control study randomized 2,856 patients and demonstrated a rate of recurrent VTE which was considered noninferior for dabigatran (1.8 % vs. 1.3 %) with significantly less frequent major or clinically relevant bleeding in patients treated with dabigatran (Schulman et al. 2013). The placebo-control study randomized 1,343 patients demonstrating a significant reduction in recurrent VTE in patients receiving dabigatran but with a significantly greater rate of major or clinically relevant bleeding. The authors conclude that dabigatran is effective for the extended treatment of VTE with a lower risk of major or clinically relevant bleeding compared to warfarin but a higher risk of bleeding compared to placebo. Interestingly, there was a higher rate of acute coronary syndrome in the dabigatran arm of the active-control study (0.9 % vs. 0.2 %, $P = 0.02$). The authors point out that this phenomenon was not seen in the placebo-control study. A similar finding was suggested in the RE-LY study (dabigatran vs. warfarin in patients with atrial fibrillation), but further evaluation of the data seemed to eliminate its significance. However, a meta-analysis of seven trials showed significantly higher risk of myocardial infarction or ACS associated with dabigatran (Uchino and Hernandez 2012).

Pharmacokinetics

Dabigatran etexilate is a small synthetic molecule administered orally as a prodrug which is converted in the liver to the active compound, dabigatran. It is a competitive reversible direct inhibitor of both free and clot bound thrombin through binding of the active site of the thrombin molecule. It has a rapid onset of action (peak plasma concentration and anticoagulant effect) which is achieved within 1.5–3 h of dosing. Its elimination half-life is 14–17 h (longer in the elderly). Dabigatran is excreted unchanged by the kidneys (80 %) and eliminated via bile (20 %). Dabigatran does not induce or inhibit cytochrome P450 drug metabolizing enzymes. It is a substrate for P-gp (permeability glycoprotein) which gives rise to clinically important drug interactions (i.e., ketoconazole and amiodarone increase plasma concentrations). Concomitant administration of strong P-gp inhibitors (i.e., itraconazole and protease inhibitors like ritonavir) with dabigatran requires great caution or is contraindicated. Moderate inhibitors of P-gp (i.e., macrolides like clarithromycin) are not considered to have a clinically important interaction. Rifampicin is a P-gp inducer causing a decreased anticoagulant effect and because dabigatran requires a low pH for absorption, proton pump inhibitors (PPI) may also decrease its anticoagulant activity. Dabigatran has low plasma protein binding (35 %) thus is not affected by displacement interactions which can occur when a second medication is administered. Food prolongs time to peak plasma levels by about 2 h (Poulsen et al. 2012).

Coagulation Assays

Because of the predictable pharmacokinetics exhibited by dabigatran routine monitoring is not necessary. However, there may be circumstances that arise in which knowledge about the anticoagulation status of a patient is needed. Situations that may arise include active bleeding, overdose and when urgent or emergent surgery is needed. Dabigatran inhibits thrombin mediated conversion of fibrinogen to fibrin therefore will affect all routine coagulation assays. The activated partial thromboplastin time (aPTT) assesses the intrinsic pathway of the coagulation cascade and prolongs as dabigatran plasma concentration increases, but this response flattens at high contraindications. The aPTT is relatively insensitive within the range of dabigatran plasma concentrations seen in patients. Therefore, it may be useful qualitatively, confirming presence of dabigatran anticoagulant activity, but it is not appropriate for quantifying anticoagulant effects, especially at high plasma concentrations. The prothrombin time (PT) assesses the clotting time in the extrinsic pathway of the coagulation cascade and is insensitive to prophylactic doses of dabigatran and elevates slightly at therapeutic doses. Therefore, it is of limited value in determining the anticoagulant status of a patient. The thrombin time (TT) is a direct measure of thrombin activity in plasma and has a linear dose response over therapeutic concentrations. However, standardized reference ranges are not possible because reagents differ among laboratories; furthermore, as dabigatran concentrations increase, the test will often exceed the maximum measurement time of coagulometers, and thus thrombin time may not be ideal (too sensitive) in the emergent setting (Stangier et al. 2007). It is most useful to determine presence of dabigatran activity rather than to quantify the level. The activated clotting time (ACT) is a quantitative assay using whole blood samples. There is a linear concentration-dependent increase in ACT up to 250 ng/ml and then the response flattens similar to that seen with aPTT testing, limiting its utility. The Hemoclot® thrombin inhibitor assay (Hyphen Biomed, France®) is a diluted TT assay allowing for quantitative measurements of DTI activity in plasma based on a constant defined concentration of thrombin. Diluted test plasma is mixed with normal pooled human plasma and clotting is stimulated by adding highly purified human thrombin. There is a direct linear relationship between dabigatran concentration and clotting time (about 30–75 s). The test can be used for any DTI (Stangier and Feuring 2012).

The ecarin clotting time (ECT) measures thrombin generation. Ecarin is isolated from snake venom (*Echis carinatus*) which activates prothrombin resulting in generation of meizothrombin (precursor of thrombin). The DTIs inhibit the thrombin-like activity of meizothrombin therefore the ECT can directly measure DTI activity. There is a linear relationship between ECT prolongation and DTI plasma concentration. The ECT has been demonstrated to be a sensitive test for measuring DTI anticoagulant activity however it is mostly used as a research tool and is not yet widely available or validated for routine clinical use in emergent settings (Van Ryn et al. 2010).

Reversal of Dabigatran

Under most circumstances, withholding drug and waiting for its activity to dissipate is adequate without the need for a specific antidote. The length of time to withhold dabigatran will depend on the patient's renal function and reason for anticoagulation activity reversal. The half-life of dabigatran in patients with normal to mildly impaired renal function (creatinine clearance above 50 ml/min) is 15 h (range 12–34 h); therefore, for standard risk elective surgery, one should hold dabigatran for 24 h prior to surgery. If the surgery or patient is considered higher risk for bleeding and complete hemostasis is desired then the drug should be held 2–4 days before surgery (Van Ryn et al. 2010). The half-life of dabigatran in patients with moderately impaired renal function (creatinine clearance 30–50 ml/min) is 18 h (range 13–23 h) and holding the drug for 2 days is suggested for standard bleeding risk procedures, and at least 4 days for higher bleeding risk patients. For patients at greater risk of bleeding, measurement of a TT 6–12 h before surgery can be performed and if elevated indicates persistence of DTI activity and the elective surgery should be postponed. Patients with severe renal dysfunction (creatinine clearance <15 ml/min) should not be prescribed dabigatran. In case of recent overdose (within 1–2 h) activated charcoal may be given. If the overdose occurred more than 2 h from the presentation, further removal of dabigatran from plasma by hemoperfusion over a charcoal filter is under investigation. If a patient on dabigatran presents with bleeding, as with any anticoagulant, the severity of bleeding, timing of last dose, degree of coagulopathy, and clinical response to supportive measures should guide medical intervention decisions. The medication should be withheld, the site of bleeding should be compressed, and surgical hemostasis should be provided if possible. Transfusion of blood products should be provided as indicated (red blood cells, fresh frozen plasma). Dabigatran plasma concentrations decline relatively quickly following discontinuation of the drug. If supportive measures are inadequate, hemodialysis may be an option, although there is limited guiding data. In cases of the need for more rapid reversal of anticoagulant activity, especially in a patient with renal impairment dialysis has been shown to remove 62 % of drug at 2 h and 68 % at 4 h (Stangier et al. 2010). There is limited and inconsistent data on the utility of recombinant coagulation factor VIIa in the reversal of anticoagulant activity from direct thrombin inhibitors (Marlu et al. 2012). Additionally, use of prothrombin complex concentrates (PCC) is under investigation. In one randomized, double-blind, placebo controlled study, dabigatran's anticoagulant effect was not reversed with PCC (Eerenberg et al. 2011).

Rivaroxaban

Rivaroxaban is a new oral anticoagulant which works chiefly by direct, selective, and reversible inhibition of factor Xa, thereby inhibiting platelet activation and fibrin clot formation. In 2011 the US FDA approved rivaroxaban for prevention of stroke and systemic embolization in patients with non-valvular atrial fibrillation and for prevention of VTE in patients undergoing knee or hip replacement surgery. In 2012 the US FDA expanded approval of rivaroxaban to include treatment of VTE

Key Clinical Trials

In a large double-blind clinical trial of 14,264 patients with nonvalvular atrial fibrillation at elevated risk for stroke with a CHADS2 score of 2 or above were randomized to either rivaroxaban (20 mg once daily) or dose-adjusted warfarin (Patel et al. 2011). Rivaroxaban was found to be noninferior to warfarin at preventing stroke or systemic embolization with no difference in the risk of major bleeding. In fact, there was significantly less intracranial hemorrhage (0.5 % vs. 0.7 %, $P=0.02$) and fatal bleeding (0.2 % vs. 0.5 %, $P=0.003$) in the rivaroxaban group (ROCKET-AF clinical trial).

In a pooled analysis of four phase 3 double-blind, randomized studies (RECORD1–4) directly comparing rivaroxaban (10 mg once daily) to enoxaparin (40 mg once daily or 30 mg twice daily) for thromboprophylaxis following hip or knee arthroplasty, including 12,729 patients, rivaroxaban had superior efficacy at reducing the composite end point of symptomatic VTE and all-cause mortality after hip or knee arthroplasty. There was no significant difference in the incidence of major bleeding between rivaroxaban and enoxaparin (Turpie et al. 2009, 2011; Eriksson et al. 2008; Kakkar et al. 2008; Lassen et al. 2008). In another phase III trial, rivaroxaban (10 mg once daily) was compared to enoxaparin (40 mg once daily) in acutely ill medical patients (Cohen et al. 2013). The patients were randomized if they were 40 years of age or older and hospitalized with an acute medical illness. Thromboprophylaxis was provided for 10 ± 4 days with either drug and an extended treatment with rivaroxaban out to 35 ± 4 days was compared to placebo. The primary efficacy outcome was VTE occurring up to 10 days for the noninferiority test and up to 35 days for the superiority test. A total of 8,101 patients underwent randomization with just under 3,000 patients being randomized to each of the two medication groups. At 10 days there was a difference in the incidence of the primary efficacy outcome. Extended therapy rivaroxaban resulted in statistically fewer VTE events compared to enoxaparin followed by placebo at 35 days. Rivaroxaban was found to be noninferior compared to enoxaparin for thromboprophylaxis in acutely ill medical patients; however, there was significantly more major bleeding associated with rivaroxaban at both 10 and 35 days.

Three large phase 3 trials have been conducted to assess for treatment of VTE (EINSTEIN-EXT), treatment of acute DVT (EINSTEIN-DVT) and acute PE (EINSTEIN-PE) with rivaroxaban. Rivaroxaban (15 mg twice daily for 3 weeks followed by 20 mg once daily) was compared to enoxaparin followed by adjusted dose warfarin for treatment of acute deep DVT in an open-label, randomized, noninferiority study. Treatment duration was 3, 6, or 12 months in patients with acute symptomatic DVT. The study included 3,449 patients, 1,731 were randomized to rivaroxaban and 1,718 were treated with warfarin. Rivaroxaban was noninferior at preventing recurrent VTE with identical rates of major bleeding or clinically relevant nonmajor bleeding. The extended treatment arm of the study was double-blind and randomized and designed to assess for superiority of continued rivaroxaban (20 mg once daily) compared to placebo for an additional 6–12 months. There were 602 patients in the rivaroxaban group and 594 in the placebo group. Superior efficacy was demonstrated with eight recurrent VTE events occurring in the treatment group compared to 42 events in the placebo group. There was no statistical difference in bleeding rates (Bauersachs et al. 2010). In a similarly designed noninferiority study, 4,832 patients with acute symptomatic PE with or without DVT were randomized to either rivaroxaban or standard therapy (enoxaparin followed by adjusted dose vitamin K antagonist) for 3, 6, or 12 months. Rivaroxaban again proved noninferior with a statistically significant lower rate of major bleeding (Buller et al. 2012).

Rivaroxaban has also been evaluated for ACS. In the ATLAS ACE-TIMI 51 study (Mega et al. 2012) 15,526 patients with non-ST elevation myocardial infarction (MI), and ST MI or unstable angina were randomized to either rivaroxaban (2.5 mg twice daily or 5 mg twice daily) or placebo along with standard therapy.

Patients were enrolled within 7 days of an event, after intervention and stabilization. The primary endpoint was a composite of cardiovascular death, MI, or stroke. Rivaroxaban was effective at reducing the primary composite endpoint at both doses. The lower dose reduced cardiovascular death and all-cause mortality but not the higher dose and there were increased rates of major bleeding and intracranial hemorrhage but not fatal bleeding (Mega et al. 2012).

Pharmacokinetics

Rivaroxaban is a reversible, direct inhibitor of not only free factor Xa but also prothrombinase complex and clot-associated factor Xa. Inhibition of factor Xa incorporated in the prothrombinase complex is a unique feature of small direct inhibitors and not a characteristic of antithrombin or antithrombin-dependent anticoagulants (Ageno et al. 2012). It is rapidly absorbed with high bioavailability and maximum plasma concentration is reached in 2–3 h. Its mean elimination half-life is 7–11 h in young healthy adults (11–13 h in the elderly). The effect of age, extreme body weight and gender on pharmacokinetic and pharmacodynamic properties of rivaroxaban has been evaluated. Findings suggest a single dose can be safely administered to all patients regardless of age, gender, or body weight. It is highly bound to plasma proteins in the range of 92–95 %, mostly albumin, and therefore it is not removable by hemodialysis. Elimination is about two-thirds via CYP3A4 and CYP2J2 and through CYP-independent mechanisms. About two-thirds are excreted by the kidneys; one-third of the active drug unchanged and one-third as inactive metabolites. The remaining third of the drug is metabolized by the liver and excreted in the feces. Drug interactions may occur with strong inhibitors for CYP3A4 and P-gp (i.e., ketoconazole and ritonavir). Rivaroxaban is not recommended in patients requiring treatment with azole-antimycotics (i.e., ketoconazole) or HIV protease inhibitors because of increased risk of bleeding. Macrolides (i.e., clarithromycin, erythromycin) are strong inhibitors of CYP3A4 and moderate P-gp inhibitors; however, they are not considered to have a clinically important interaction with rivaroxaban. Rifampicin is an inducer of CYP3A4 and P-gp and therefore will significantly decrease the effectiveness of rivaroxaban (Poulsen et al. 2012; Ageno et al. 2012).

Coagulation Assays

Rivaroxaban prolongs prothrombin time (PT), activated partial thromboplastin time (aPTT), and the *HEPTEST*. It does not affect bleeding time or platelet aggregation and has no effect on thrombin or antithrombin activity. Inhibition of factor Xa activity and PT prolongation correlates strongly with plasma concentrations in a linear fashion. These predictable properties are what allow for fixed dosing without the need for laboratory monitoring. However, there will be circumstances when measuring the anticoagulant effect is needed such as in cases of accidental overdose, bleeding, to assess compliance, to evaluate effect of drug interactions, and to evaluate for drug accumulation in patients with renal or liver impairment. Despite having predictable effects on multiple coagulation laboratory tests, there is presently no validated laboratory assay which can be used quantitatively to monitor rivaroxaban and there are no therapeutic targets or suggested dose adjustments recommended. The effect of rivaroxaban on the aPTT is weak and therefore the test is not suitable for quantitative monitoring. Standard methods for the HEPTEST, which measures factor Xa inhibition directly proportional to the concentration of heparin present, and prothrombinase-induced clotting time, sensitive to factor Xa and IIa inhibitors, gives paradoxical results. Assays used to measure anti-Xa activity for heparin and low-molecular-weight heparins show a dose-dependent but variable response (Samama et al. 2012). More validation and further study are needed to develop a reliable quantitative assay for clinical use (Ageno et al. 2012).

Reversal of Rivaroxaban

No specific antidote is yet available for reversal of the anticoagulant effects of rivaroxaban. In case of overdose or active bleeding the drug should be

discontinued and activated charcoal may be given to reduce absorption if the ingestion was recent. Dialysis is not likely to be helpful because of high plasma protein binding. Efficacy of using a particular hemostatic blood product has not yet been established. High dose four-factor PCC has been shown to normalize bleeding time in rats treated with intravenous rivaroxaban (Ageno et al. 2012) and to reverse its anticoagulant effect in healthy subjects (Eerenberg et al. 2011). Recombinant factor VIIa is suggested in cases of life-threatening bleeding however efficacy is not conclusively established (Ageno et al. 2012).

Apixaban

Apixaban is an oral direct inhibitor of factor Xa approved by the US FDA in January 2013 for prevention of stroke and systemic embolization in patients with non-valvular atrial fibrillation.

Key Clinical Trials

In a large randomized, double-blind noninferiority trial (ARISTOTLE) apixaban (5 mg twice daily) was compared to adjusted dose warfarin in 18,201 patients with atrial fibrillation with at least one additional risk factor for stroke. The primary outcome was ischemic or hemorrhagic stroke or systemic embolization. A secondary objective was to assess for superiority for primary outcome, major bleeding and death from any cause. Patients were followed for a median of 1.8 years with a significantly lower rate of the primary outcome (1.27 % per year with apixaban vs. 1.60 % per year with warfarin) demonstrating both noninferiority and superiority ($P<0.001$, $P=0.01$ respectively). Apixaban appeared to cause less bleeding and was associated with lower mortality compared to warfarin (Granger et al. 2011). In the AVERROES study, apixaban was compared to aspirin in patients with higher than average risk of thromboembolism and atrial fibrillation that were reluctant to take warfarin or thought to be unsuitable candidates. The study was double-dummy designed and 5,599 patients were randomized to either apixaban (5 mg twice daily) or aspirin (81–324 mg daily). The study terminated early because the primary end point of stroke or systemic embolism was 1.6 % in the apixaban group and 3.7 % in the aspirin group while rates of major bleeding and intracranial hemorrhage were similar (Connolly et al. 2011).

In the APPRAISE-2 study apixaban was assessed for the treatment of ACS. There was no significant difference in primary endpoint of cardiovascular death, MI or ischemic stroke in patients receiving standard therapy plus apixaban however there was an increased rate of major bleeding (1.3 % on apixaban vs. 0.5 % on placebo). The study was therefore terminated early (Alexander et al. 2011).

There have been three large double-blind, randomized, phase 3 clinical trials looking at apixaban for prophylaxis following major joint replacement surgery (ADVANCE-1-3). In the first study, 3,195 patients were randomized to either apixaban 2.5 mg twice daily or enoxaparin 30 mg subcutaneously injected every 12 h starting 12–24 h after total knee replacement surgery and continuing on for 10–14 days. The primary outcome defined as a composite of asymptomatic and symptomatic VTE and death from any cause during treatment was 9.0 % for the apixaban group and 8.8 % for the enoxaparin group. The rate of VTE in both groups was lower than expected which led to incorrect assumptions when determining the necessary sample size and criteria for calculating noninferiority. As a consequence, the criteria for noninferiority were not met for the primary outcome. However, the rate of major bleeding combined with clinically relevant nonmajor bleeding was 2.9 % in the apixaban group and 4.3 % in the enoxaparin group. Although apixaban did not meet the predetermined criteria for inferiority in this study, the authors conclude that it appears to have similar efficacy to enoxaparin with a lower risk for hemorrhagic complications (Lassen et al. 2009). In ADVANCE-2 apixaban 2.5 mg twice daily is again studied in total knee replacement surgery patients compared to enoxaparin with a slightly different dosing regimen. Enoxaparin was administered as a 40 mg subcutaneous injection once every 24 h beginning 12 h before surgery and

continuing for 10–14 days postoperatively. The primary endpoint was a composite of asymptomatic and symptomatic VTE and all cause death during treatment. A total of 3,057 patients were randomized to either treatment arm. The primary outcome occurred in 15 % of the apixaban group versus 24 % of the enoxaparin group translating to an absolute risk reduction of 9.3 %. Major or clinically relevant nonmajor bleeding occurred similarly in both groups. The authors conclude not only that apixaban is noninferior to this dosing regimen of enoxaparin, but that it is more effective for prevention of venous thromboembolism without increasing the risk of hemorrhagic complications (Lassen et al. 2010a, b). Total hip replacement surgery patients have also been studied using apixaban versus enoxaparin (ADVANCE-3). Apixaban 2.5 mg twice daily was compared with enoxaparin 40 mg once daily for thromboprophylaxis in patients following hip arthroplasty. This double-blind, double-dummy study enrolled 5,407 hip surgery patients. Thromboprophylaxis was extended to 35 days and the primary end point was a composite of asymptomatic or symptomatic DVT, nonfatal PE, or any-cause mortality. Apixaban met criteria for both noninferiority and superiority for the primary endpoint with similar rates of major bleeding (Lassen et al. 2010a, b).

In the ADOPT trial extended thromboprophylaxis with apixaban beyond hospital discharge was studied in medically ill hospitalized patients to determine if it could reduce the risk of VTE better and safer than using short term enoxaparin. In this double-blind, double-dummy, placebo controlled trial 4,995 patients were randomized and evaluated for the primary efficacy outcome. Eligible patients were hospitalized with congestive heart failure, respiratory failure, or other medical disorders with at least one additional risk factor for VTE and an expected hospitalization of at least 3 days. Patients were randomized to either apixaban 2.5 mg twice daily for 30 days or enoxaparin 40 mg once daily for 6–14 days. The primary efficacy outcome was a composite of death related to VTE, PE, symptomatic VTE, and asymptomatic proximal DVT at 30 days. Bleeding was the primary safety outcome. There was a 13 % reduction in the primary efficacy outcome in the apixaban group however the difference was not statistically significant (2.71 % in the apixaban group, 3.06 % in the enoxaparin group, $P = 0.44$). Major Bleeding was higher in the apixaban group (0.47 % vs. 0.19 %, $P = 0.04$). The authors concluded that extended thromboprophylaxis with apixaban was not superior to short course enoxaparin in medically ill patients and that apixaban was associated with more major bleeding events (Goldhaber et al. 2011). It should be noted that the rate of primary efficacy outcomes immediately increased once the enoxaparin injections were stopped and that extending thromboprophylaxis with enoxaparin beyond hospital discharge in medically ill patients is not usual medical practice. Therefore, there may in fact be a benefit of extending pharmacologic thromboprophylaxis in medically ill patients and further study and refinement of patient selection may clarify this in the future.

Apixaban is being studied for the treatment of VTE in an ongoing double-blind, phase 3 efficacy and safety clinical trial (AMPLIFY). It is being compared with enoxaparin to adjusted dose warfarin for a duration of 6 months for the treatment of acute VTE. A total of 4,816 patients are enrolled. The study is now closed to enrollment and final data collection results are not yet available. Apixaban has also been studied as an option for extended anticoagulant therapy for VTE in patients whom there was clinical equipoise in terms of continuing versus stopping anticoagulants (AMPLIFY-EX). The study was double-blinded and randomized 2,486 patients to receive either apixaban 2.5 mg twice daily or 5 mg twice daily compared to placebo for a duration of 12 months. All patients had already received a 6–12 month course of anticoagulation for the initial acute VTE. The primary endpoint of symptomatic recurrent VTE or death was much higher in the placebo group (8.8 %) as compared to either the prophylactic dose (1.7 %) or treatment dose (1.7 %) apixaban groups. There was no significant difference

in rates of major bleeding among all three groups however there was a trend toward increased clinically relevant nonmajor bleeding in the apixaban groups. The investigators concluded that extended anticoagulation with either prophylactic or treatment doses of apixaban results in a reduction of recurrent VTE while not significantly increasing the risk of major hemorrhage (Agnelli et al. 2013).

Pharmacokinetics

Like rivaroxaban, apixaban is an oral, direct, reversible inhibitor of factor Xa. It is absorbed rapidly and has no food interactions. The onset of action is 2–3 h and it is excreted 25 % by the kidneys and 75 % by the hepatobilliary route. The elimination half-life is 8–15 h in healthy adults. It is metabolized mostly by CYP3A4, CYP3A5, and sulfotransferase 1A1. Half of the dose is excreted unchanged in the feces. It is a substrate for P-gp. Anticoagulant effects are significantly increased by coadministration with strong inhibitors of both CYP3A4 and P-gp (i.e., ketoconazole, HIV protease inhibitors). Significant decreased plasma concentrations are seen with potent inducers of CYP3A4 and P-gp, most importantly rifampin. Phenytoin and carbamazepine are also strong inducers and would likely have a similar effect on plasma concentrations (Poulsen et al. 2012).

Coagulation Assays

Like the other oral directly inhibiting anticoagulants, apixaban does not require routine monitoring. Similar to rivaroxaban the PT, INR and aPTT are prolonged with apixaban, however the elevations seen in these clotting tests are often very small, even at therapeutic doses and variable results will be obtained depending on which reagent is used by a particular lab. This means a patient could be fully anticoagulated while their protime and aPTT are only slightly elevated. While not yet routinely available, an anti-Xa chromogenic assay calibrated using an apixaban standard has been shown to correlate with apixaban plasma concentrations (Rojas-Hernandez and Garcia 2013). There are no established therapeutic targets yet, but this test is a promising candidate for a simple and quick way to asses a patients level of anticoagulation while on factor Xa inhibitors.

Reversal of Apixaban

Similar to rivaroxaban, there is no antidote for reversal of the anticoagulant effect of apixaban. In the setting of bleeding or need for reversal the drug should be discontinued and activated charcoal can be used to reduce absorption of the medication if ingestion of the last dose was recent. The anticoagulant effect will last about 24 h after the last dose (two half-lives). It has high plasma protein binding therefore dialysis will likely have little effect on plasma concentrations. Protamine sulfate and vitamin K are not useful in reversal of anticoagulant effects from apixaban. Prothrombin complex concentrate, activated prothrombin complex concentrate and recombinant factor VIIa may be considered in the setting of life threatening bleeding however it has not been studied in clinical trials (source, prescribing highlights).

Future Oral Anticoagulants

There are numerous additional oral anticoagulants being studied at various stages of investigation. The most common target is factor Xa. Additional targets include thrombin, factor IXa, and factor XIa (Al-Horani et al. 2013). The most common indication being evaluated is for thromboprophylaxis after hip or knee replacement surgery. Additionally, factor Xa inhibitors betrixaban and eribaxaban are being studied for prevention of embolic complications of atrial fibrillation and letaxaban is being studied in patients with ACS (Ahrens et al. 2012) (Table 16.1). A summary of the features of the currently approved US FDA new anticoagulants is found in Table 16.1.

Table 16.1 Features of the new oral anticoagulants

	Dabigatran etexilate	Rivaroxaban	Apixaban
Target	Thrombin	Factor Xa	Factor Xa
Form	75 and 150 mg capsules	10, 15, 20 mg tablets	2.5 and 5 mg coated tablets
FDA approved indications	Non-valvular a-fib	Non-valvular a-fib, VTE prophylaxis after hip/knee replacement, treatment of VTE	Non-valvular a-fib
Dosing	150 mg BID if the CrCL is >30 ml/min and 75 mg BID if the CrCl is 15–30 ml/min	a-fib: 20 mg QD or 15 mg QD if CrCl is 15–50 ml/min VTE treatment: 15 mg BID × 21 days then 20 mg QD VTE prevention after hip/knee replacement: 10 mg QD	5 mg BID In patients with at least 2 of the following characteristics: age ≥80 year, body weight ≤60 kg, serum creatinine ≥1.5 mg/dL, the recommended dose is 2.5 mg orally BID
Bioavailability	3–7 %	~80 %	>50 %
Food	Delays absorption	No affect	No affect
Maximum concentration	1.5 to 3 h	2–3 h	3–4 h
Half-life	14–17 h	7–11 h	8–15 h
Elimination	80 % renal, 20 % liver	66 % renal, 33 % feces	~25 renal, 50 % biliary
Drug interactions: CYP/P-gp	No interaction with CYP, substrate for P-gp	3A4, 2J2, substrate for P-gp	3A4, substrate for P-gp
Renal impairment	CrCl < 30 reduce dose CrCl < 15 or on HD contraindicated	CrCl < 50 reduce dose CrCl < 30 or on HD contraindicated	CrCl < 15 or on HD contraindicated
Liver impairment	Contraindicated in severe liver impairment	Contraindicated in severe liver impairment	Contraindicated in severe liver impairment
Pregnancy/lactation	Insufficient data	Insufficient data	Insufficient data

FDA Food and Drug Administration, *a-fib* atrial fibrillation, *VTE* venous thromboembolic disease, *BID* twice daily, *QD* once daily, *CYP* cytochrome P450, *P-gp* P-glycoprotein

Discussion of Clinical Vignettes

Clinical Vignette 1 discussion: None of the new oral anticoagulants have been well studied in pregnancy or in lactating woman. They are all contraindicated in this setting.

Clinical Vignette 2 discussion: An advantage of the new oral anticoagulants is improved convenience and less complexity. However, they do not improve compliance with taking the medication. In the clinical setting of poor compliance, patients require close monitoring and more frequent visits to reinforce the need to take their medication. Dabigatran would not be a good option for this patient.

Clinical Vignette 3 discussion: The half-life of dabigatran in patients with normal to mildly impaired renal function is about 15 h thus for a standard risk elective surgery 24 h should be adequate for holding the drug. If bleeding risk is felt to be greater, then holding longer is advised (2–4 days). The most readily available and sensitive test to assess for the presence of direct thrombin inhibitor (DTI) activity is the thrombin time (TT). This test can be run 6–12 h prior to surgery and if elevated indicates DTI activity persists and the surgery should be postponed until it normalizes.

(continued)

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